Interpretable Used-Tesla Asking Prices: Trim Indicators and Model-Specific Registration Effects

G01

Abstract

Buyers and small dealers need comparable asking-price estimates when screening used electric vehicles. We study 3,754 European Tesla advertisements using a supplied 3,002/752 training/test partition. Ordinary least squares predicts price in euros from seven structured inputs, six title-derived trim indicators, and three model-by-registration-year interactions. The engineered model reduces dealer-grouped cross-validation MAE from EUR 3,991 to EUR 3,324, and supplied-test MAE from EUR 4,290 to EUR 3,711. A random forest remains better at EUR 3,227. Neither meets our proposed EUR 3,000 test-MAE threshold. Ablations isolate the value of each engineered group, and coefficient and case analyses expose sparse-category uncertainty and extreme-listing failures. Because the test set was inspected during earlier development, these are descriptive benchmark results, not a new blind evaluation.

1 Introduction

A Tesla model name alone does not identify its trim, software entitlement, or condition. A buyer screening advertisements needs a comparable-price reference before paying for inspection; a small dealer needs an initial asking-price check. We estimate the *advertised* price available in the observed market, not a sale price, future value, or certified fair value. An unusually large price gap should trigger inspection, not an automatic purchase decision.

Our question is whether inexpensive, listing-time trim indicators and model-specific registration-year slopes improve a transparent linear estimator beyond raw attributes and correlation ranking. We contribute (1) a fixed-split, train-only feature analysis (Sections 3–5), (2) matched linear ablations with dealer-grouped validation (Sections 6–7), and (3) final-model uncertainty and failure-case analysis (Section 8). The engineered model remains linear in its coefficients; we make no claim that it beats the nonlinear baseline.

2 Related Work

Pudaruth (2014) retained 97 Mauritian newspaper advertisements, using year and cylinder volume for numerical prediction; sparse mileage limited its use. Their 10-fold kNN MAE was 45,189 rupees for Toyota ($k = 1$) and 25,741 for Nissan ($k = 5$). We instead use Tesla listings across European seller countries, with explicit trim and usage features.

Bukvić et al. (2022) studied Croatian Njuškalo advertisements, with manufacturer/model, production year and kilometres as candidate attributes. Their Table 3 reports linear-regression MAE 1,373.2 and RMSE 2,104.7 on the EUR price scale, versus random forest and SVM. We focus on one electric-vehicle brand and distinguish first registration from manufacture.

Madhusudhanan et al. (2024) use about two million B2C records and 65 attributes, including age, mileage and offer duration. Their attention-based ProbSAINT reports MAPE 5.6% and 5.3% on two temporal splits (Table 2). Our small snapshot lacks sales outcomes and duration; we use only listing-time fields. Different populations, currencies and targets make these errors context, not head-to-head benchmarks.

3 Task Definition and Data

One row is an AutoScout24 advertisement. The continuous target is its asking price in EUR. The training range is EUR 9,000–175,450; it defines observed coverage, not a clipping rule. Inputs are first-registration year, mileage (km), reported power (kW), equipment count, model, seller country, assistance label and selected title keywords.

All are available when screening a listing with its price hidden. Seller identity, URL, collection order and target-bearing free text are not predictors.

The supplied workbook records collection on 5 October 2026 across 188 result pages in eight European countries. It contains 3,754 listings, with the first 3,002 in training and 752 in testing. We retain this original partition for reproducibility. Search-page order is neither randomized sampling nor longitudinal forecasting; a later-market claim would require a temporal dataset. The collector's listing-ID deduplication does not establish unique physical vehicles.

Variable	Mean	SD	Range
Price (EUR)	35,308	13,937	9,000–175,450
Year	2021.46	2.43	2001–2026
Mileage (km)	88,835	58,372	9–635,000
Power (kW)	300	126	7–1,020
Equipment	55	22	2–119

Table 1: Training-only summary ($n = 3,002$). Inherited fields retain suspect values; range is observed, not a validity bound.

All seven structured inputs are populated in the supplied workbook, so no additional imputation is fitted. Earlier filling and assistance annotations are inherited, not independently verified. Numeric fields are standardized using the current training fold; categories are one-hot encoded with unknown categories ignored at prediction. We preserve price extremes, because unusually expensive cars need not be erroneous. One training Model Y has year 2001: the main benchmark retains it to match the shared data, and Section 7 reports an exclusion sensitivity. Reported power and equipment counts are seller fields, not certified specifications; equipment lists can include headings. Appearance and range are excluded from the primary model to avoid adding uncertain measurements without demonstrated incremental value.

4 Exploratory Feature Analysis

Figure 1 uses training rows only. Price is associated with mileage, year and vehicle segment, but each marginal plot mixes trims and countries. Training Pearson correlations with price are -0.403 for mileage, 0.286 for year and 0.557 for power. The year–mileage correlation is -0.594; this motivates examining redundancy rather than interpreting either slope in isolation. We retain power as a predictive seller annotation, while acknowledging its uncertain measurement basis.

Univariate numeric r^2 and categorical correlation ratio η^2 place power, model, mileage, year and country above equipment and assistance. These common-scale scores use raw EUR price, not log-price. We test three explicit hypotheses: **H1**, trim indicators reduce error conditional on structured fields; **H2**, model-specific year interactions improve on a shared slope; **H3**, these two groups outperform raw correlation ranking. Their verdicts come from matched ablations in Section 7 and coefficient interpretation in Section 8, not from these plots alone.

5 Method

5.1 Pipeline and feature groups

Figure 2 traces each listing from raw fields to a EUR prediction. **Set A** contains the seven structured fields. **Set B** retains the top k fields ranked inside each training fold: squared Pearson correlation for numeric fields and between-category/total target sum of squares for nominal fields. Searching $k = 1, \ldots, 7$, the smallest value within one standard error ($s/\sqrt{5}$) of the lowest CV MAE is $k = 5$, favoring a smaller representation when gains are within fold uncertainty.

Set C, our final OLS representation, keeps all seven raw fields and adds two groups. Six case-insensitive title flags mark Performance/Plaid, Long Range, AWD/Dual Motor, entry trim (RWD/Standard Range), new generation (Highland/Juniper/Plaid/refresh) and HW4. These capture within-model differences that power alone may miss (H1). Flags are mentions, not verified configurations: Plaid activates two flags, and Standard Range does not certify rear-wheel drive. The reproducibility package preserves exact expressions.

Three interactions allow Model S, X and Y to have different registration slopes:

$$u_m = \mathbf{1}(\text{model} = m)(\text{year} - \overline{\text{year}}_{\text{fit}}). \quad (1)$$

They address H2. The fit-partition mean prevents validation-dependent centering. Model 3 shares the base slope with rare models; these latter classes cannot support reliable separate slopes. This is a registration-year association, not observed longitudinal depreciation.

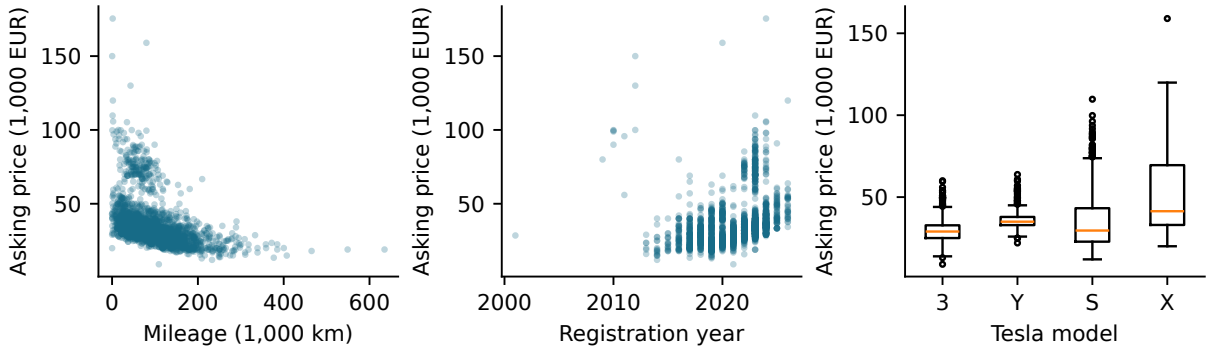


Figure 1: Training-only price against mileage, first-registration year and Tesla model ($n = 3,002$). Units are shown on every axis. Scatterplots retain extremes; boxes show quartiles, with 1.5-IQR whiskers. Association is not a causal depreciation estimate.

Advertisement: structured fields + title
↓
Parse six keyword flags; form three model × year terms
↓
Training-fold scaling + category vocabulary
↓
OLS: standardized representation → EUR price
↓
Fixed validation/test rows → MAE, RMSE, R^2

Figure 2: Prediction pipeline. Every learned transformation uses only its fit partition. The target enters regression fitting and evaluation, never feature construction.

5.2 Linear model and alternatives

OLS minimizes $\sum_i (y_i - b - \mathbf{z}_i^T \boldsymbol{\beta})^2$. It offers direct EUR effects and a small fixed representation, while interactions allow non-additive raw-input behavior. It has no regularization parameter. No target logarithm, post-hoc clipping or higher-order polynomial search is used. Full one-hot fitting uses a least-squares solution; inference uses an equivalent full-rank reference coding. The retained representation has 29 coefficients including the intercept.

Ridge is a plausible alternative for correlated features. A training-only dealer-grouped sensitivity searches $\alpha \in \{.01, .1, 1, 10, 100, 1000\}$ on the same Set C inputs; the lowest MAE is EUR 3,324 at $\alpha = 0.01$, compared with OLS EUR 3,324. We retain the already specified OLS benchmark and report this alternative without choosing a new model from test outcomes. Coefficient uncertainty is conditional on this representation.

6 Experimental Setup

6.1 Validation and provenance

We use identical five-fold shuffled KFold partitions (seed 42) across methods, plus five-fold GroupKFold by seller/dealer as the structure-aware check. Five folds retain about 80% for fitting while permitting a spread estimate at modest cost. There are 1,749 training dealer groups; each missing seller receives its own group. Scaling, encoding, centering and top-k ranking are fitted within folds. We report mean ± sample SD. Feature-set/k exploration used these training folds, so CV is development evidence, not a nested estimate of the entire selection process.

The supplied test set was already inspected in earlier team analyses. We freeze this integration's model definitions before recomputation and retain all 752 rows, but do not claim a new one-use blind holdout. Dealer grouping is not a vehicle-identity guarantee; overlap and sensitivity checks are reported separately. A newly collected locked test set remains necessary for confirmatory deployment claims.

6.2 Metrics and baselines

MAE measures typical absolute EUR error; RMSE penalizes large misses, and R^2 describes explained variation. We propose MAE $\leq$ EUR 3,000 as a screening-use threshold, not a validated stakeholder tolerance or an individual guarantee. We also report the proportion of cars within EUR 3,000.

Baselines are the training mean, mileage-only OLS, and a 200-tree random forest on Set A (seed

42; no depth limit, default leaf size 1). The forest is an inherited fixed flexible comparator, not a tuned optimum. All methods use identical rows and metrics. Paired test-loss gaps use 10,000 dealer-cluster bootstrap resamples (seed 42), keeping both methods' predictions together. Grouped-fold paired t-tests have only five overlapping training fits, so their p-values are exploratory.

7 Results

7.1 Main comparison

Method	Group MAE	Test MAE	RMSE	R^2
Mean	8,552 ± 1,354	8,906	14,040	-0.004
Mileage	7,806 ± 1,171	8,116	12,909	0.152
RF (raw)	**2,660 ± 236**	**3,227**	**5,103**	**0.867**
OLS A	3,991 ± 429	4,290	5,971	0.818
OLS C (ours)	3,324 ± 267	3,711	5,399	0.852

Table 2: Main results: errors in EUR; Group MAE is five-fold mean ± sample SD. RMSE and R^2 use all 752 test rows. Bold marks the best values.

Table 2 shows that engineered OLS improves over raw OLS but remains behind the random forest. Test MAE falls by EUR 578 (13.5%) from A to C; the forest's advantage over C is EUR 484. A piecewise tree ensemble can capture local trim/usage combinations beyond the three slopes and six flags, although the comparison does not isolate that mechanism. The forest remains a baseline; our final predictor is linear.

Against the proposed threshold, C exceeds EUR 3,000 by EUR 711, and RF exceeds it by EUR 227. Only 55.6% of C's test errors lie within EUR 3,000 (RF: 59.4%). The average threshold is not satisfied even though many individual estimates fall inside it. Thus these predictions are inspection aids, not autonomous pricing decisions.

7.2 Ablations and uncertainty

Set	KFold MAE	Group MAE	Test MAE
A: raw	3,939 ± 105	3,991 ± 429	4,290
B: top-5	3,966 ± 118	4,022 ± 446	4,329
C: full	3,244 ± 92	3,324 ± 267	3,711
C − flags	3,457 ± 95	3,522 ± 278	3,871
C − interactions	3,705 ± 58	3,779 ± 414	4,081

Table 3: Matched OLS ablations, EUR. All folds and preprocessing rules are shared. Lower is better; C is best in every column.

Only the feature representation changes in Table 3; every row uses OLS and the same splits. Removing either engineered group worsens dealer-grouped CV, supporting H1 and H2. Set C also improves on top-k Set B, supporting H3: ranking raw associations cannot create title information or model-specific slopes. Both groups use already collected fields, adding nine numeric terms rather than a new data source. Their costs are parsing maintenance and harder interpretation of overlapping trim mentions.

Comparison	Gap	95% cluster CI	p
A − C	578	[374, 796]	0.001
B − C	617	[407, 844]	0.002
RF − C	-484	[-752, -205]	0.001

Table 4: Test-MAE gaps (EUR): positive favors C. CI: paired dealer-cluster bootstrap, 10,000 draws; p: exploratory paired grouped-fold t-test.

The paired bootstrap intervals in Table 4 preserve within-dealer dependence in test losses. They describe uncertainty for these fixed fitted models and observed sellers; they omit retraining, feature selection and market change. Reused test data and uncertain vehicle identity prevent treating these intervals as definitive prospective evidence.

Model	n	A	B	C
3	347	3,467	3,481	3,004
S	69	6,790	6,917	6,289
Y	271	3,723	3,770	3,161
X	59	7,816	7,876	6,423
Roadster	5	12,403	12,395	11,976
Cybertruck	1	22,051	21,858	19,203

Table 5: Test MAE by model (EUR). All observed test slices are shown; small n limits rare-model interpretation.

Table 5 shows how gains vary by model. Larger absolute errors for S/X can reflect heterogeneous trims and price scales, not simply worse percentage accuracy. Rare classes must not be interpreted from their tiny sample sizes. The same representation can improve common-model averages while remaining unreliable on individual outliers.

Sensitivity to data quality. A model/registration-month/mileage duplicate proxy flags 152 test rows overlapping training. On the 600 remaining rows, MAE is EUR 4,550 (A), EUR 4,031 (C), and EUR 3,816 (RF). Matching attributes do not prove identical vehicles; exclusion is a sensitivity analysis, not a replacement split. Removing the single Model Y/2001

training record and refitting C gives test MAE EUR 3,713. This post-hoc audit is not used to optimize the headline model.

8 Analysis

8.1 Coefficients and significance

Term	β	95% CI	p	VIF
Year	4.56	[4.01, 5.12]	$< .001$	5.9
Mileage	-3.44	[-3.90, -2.98]	$< .001$	1.7
Power	2.78	[2.31, 3.25]	$< .001$	3.4
Equipment	0.19	[-0.03, 0.41]	0.097	1.3
Performance	1.19	[0.88, 1.50]	$< .001$	2.1
Long range	0.53	[0.18, 0.87]	0.003	2.0
AWD	-0.73	[-1.14, -0.32]	$< .001$	2.4
Entry trim	-0.61	[-0.91, -0.30]	$< .001$	2.9
New gen.	1.02	[0.76, 1.29]	$< .001$	1.4
HW4	0.74	[0.47, 1.01]	$< .001$	1.1
Model S×year	3.17	[2.61, 3.74]	$< .001$	4.2
Model X×year	2.96	[2.34, 3.58]	$< .001$	2.4
Model Y×year	-0.27	[-1.54, 0.99]	0.675	2.7
Cybertruck	133.40	—	—	1.0
Model S	20.31	[18.81, 21.81]	$< .001$	3.5
Model X	24.76	[22.81, 26.71]	$< .001$	2.5
Model Y	3.24	[1.24, 5.24]	0.002	2.3
Roadster	85.02	[65.55, 104.50]	$< .001$	1.5
Unknown	-11.69	[-28.99, 5.61]	0.185	1.0
AT	-0.09	[-0.77, 0.59]	0.790	1.3
DE	-0.39	[-1.02, 0.24]	0.221	1.7
BE	-0.84	[-1.59, -0.09]	0.028	1.3
FR	-0.29	[-1.23, 0.66]	0.550	1.2
LU	-0.10	[-3.19, 3.00]	0.950	1.0
IT	2.78	[2.08, 3.47]	$< .001$	1.5
ES	-1.43	[-2.23, -0.63]	$< .001$	1.3
EAP	0.60	[-0.35, 1.54]	0.214	1.0
FSD	0.87	[0.13, 1.61]	0.022	1.1

Table 6: Final Set C coefficients and HC3 intervals in 1,000 EUR. Standardized numeric terms, reference-coded category contrasts; all 28 non-intercept terms shown. Cybertruck ($n = 1$): uncertainty unavailable. Model contrasts apply at the training mean year.

Table 6 covers all non-intercept terms in the final Set C model, including rare categories. Numeric inputs, binary flags and interaction products are standardized before fitting; their effects are EUR per training SD. Category contrasts use Model 3, Netherlands and AP as references and remain 0/1. For a flag, divide its standardized coefficient by its SD to obtain a 0-to-1 premium. No coefficient is a percentage change, because the target is not logged.

Intervals and two-sided p-values use HC3 heteroskedasticity-robust OLS t inference. The single training Cybertruck has unit leverage: its own coefficient's interval is not identifiable. We report its fitted effect but mark interval and p unavailable; other intervals come from the algebraically equivalent fit excluding that singleton row and dummy. This avoids presenting a numerical artifact as evidence. HC3 does not correct dealer dependence; inferential results are conditional, exploratory associations, with no multiple-testing correction. VIF uses the full-rank design including an intercept.

At the training mean registration year, Model X and S contrasts are EUR 24,759 and 20,308 relative to Model 3 (Table 6). The common year slope is EUR 1,882 per year newer; S and X add 2,296 and 3,112 EUR/year. This supports heterogeneous slopes, not a causal effect of aging. The Model Y interaction interval crosses zero, so the group gain does not establish a distinct Y slope.

Mileage contributes about EUR -590 per additional 10,000 km. A Performance mention corresponds to about EUR 2,941 holding the other terms fixed. AWD has a negative conditional coefficient despite its intuitive premium: it overlaps power, model and trim flags, and is a seller-text indicator. VIF reaches 5.9, so correlated design and omitted trim details are plausible explanations. Equipment and EAP intervals cross zero; retaining them preserves the matched raw-feature baseline, not a claim that they are individually essential.

Resolving the hypotheses. H1 and H2 are predictive group hypotheses, so the relevant tests are removals in Table 3, not whether every flag or interaction separately has $p < .05$. H3 is resolved by C versus B in Tables 3 and 4. A small p-value does not make an effect useful: compare its natural-unit size to the EUR 3,000 screening tolerance. Conversely, a sparse feature can have a wide interval while the combined group improves validation MAE. Shared trim words, model mix and omitted battery condition offer alternative explanations to causal premiums.

8.2 Error analysis

Case	Actual	Predicted	Error
1	75,900	34,842	41,058
2	90,000	61,820	28,180
3	12,000	-15,506	27,506

Table 7: Three largest absolute errors of final Set C (EUR). Error = actual − predicted; context and explanations below.

Case 1: Model X 75D Dual Motor, France, 2018, 49,931 km, 193 kW. The EUR 75,900 ad-

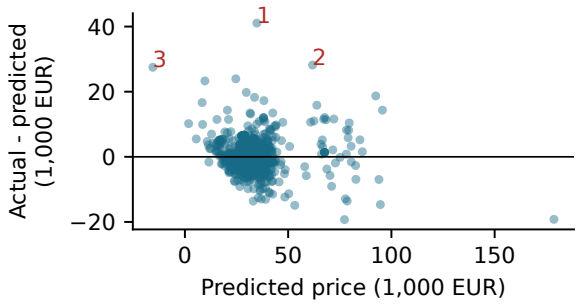


Figure 3: Set C test residuals, actual minus predicted EUR. Labels mark the three largest absolute errors in Table 7; all 752 test rows are shown.

vertisement is far above its estimate. A training-only comparison finds 27 comparable 2017–2019 Model X "75 training listings with median EUR 27,900 and maximum EUR 39,999. Unobserved condition or unusual pricing is plausible; an error cannot prove mispricing.

Case 2: Model X "75 kWh Dual Motor, Belgium, 2023, 32,000 km, 244 kW. The high asking price and registration year conflict with the older-looking title. Incorrect metadata or an unobserved configuration could explain the miss; neither is independently verified.

Case 3: Model S 85 Base, Netherlands, 2014, 670,000 km, 300 kW. Mileage exceeds the training maximum of 635,000 km. The linear extrapolation produces EUR $-15,506$, despite an actual EUR 12,000. This is a model failure, not just noisy labels. C produces one negative test prediction. The shared underprediction pattern in Figure 3 motivates input-consistency and coverage flags first; validated battery condition and a new held-out evaluation should precede any automatic correction.

9 Conclusion and Limitations

Title indicators and model-specific registration slopes improve this linear benchmark: test MAE is EUR 3,711, versus EUR 4,290 for raw OLS and EUR 3,227 for RF. Ablations support both engineered groups and improvement over raw correlation ranking. The gain is useful but insufficient for the proposed EUR 3,000 average-error threshold. Buyers should use the estimate to prioritize inspections and verify trim, battery condition and anomalous specifications before negotiating.

Coverage is one website, eight countries and one snapshot of asking prices. The supplied order split, prior test exposure, possible duplicate cars, inherited annotations, rare classes and uncertain seller measurements limit generalization. The threshold is an assumed fixed EUR cost and is relatively stricter on cheaper cars. Linear extrapolation can be implausible; fitted associations are not causal depreciation. The next experiment should freeze these rules, collect a new locked snapshot with vehicle-identity and battery-condition checks, and evaluate both price accuracy and actual screening decisions.

Contributions of Each Member

Responsibilities were shared within each subgroup. **A**: abstract, problem formulation, literature, data collection/preprocessing, EDA and feature pipeline (Sections 1–5.1; Figures 1–2). **B**: model choice/tuning, validation, metrics, threshold, baselines and main comparisons with significance testing (Sections 5.2–7.1). **C**: ablations, coefficients/significance/VIF, error cases and conclusions (Sections 7.2–9; Figure 3).

- Member 1: Group A responsibilities above.
- Member 2: Group B responsibilities above.
- Member 3: Group B responsibilities above.
- Member 4: Group A responsibilities above.
- Member 5: Group C responsibilities above.
- Member 6: Group C responsibilities above.
- Member 7: Group B responsibilities above.

AI usage. ChatGPT/Codex assisted with code revision, experimental recomputation, plots, literature checking, writing and compilation. Team packages supplied the data and feature definitions; the team retains responsibility for verifying the integrated results.

References

Lucija Bukvić, Jasmina Pašagić Škrinjar, Tomislav Fratrović, and Borna Abramović. 2022. Price prediction and classification of used-vehicles using supervised machine learning. *Sustainability*, 14(24):17034.

Kiran Madhusudhanan, Gunnar Behrens, Maximilian Stubbemann, and Lars Schmidt-Thieme. 2024. ProbSAINT: Probabilistic tabular regression for used car pricing. *Preprint*, arXiv:2403.03812. Results cited from version 1, Table 2.

Sameerchand Pudaruth. 2014. Predicting the price of used cars using machine learning techniques. *International Journal of Information and Computation Technology*, 4(7):753–764.